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Project Report

Prepared for: ACS Wil Program

Project focus: Actionable churn intelligence, customer segmentation, and retention planning

Dataset: 7,043 customer records

Document status: Final project deliverable Report

Executive Summary

Customer churn is a material commercial risk: acquiring a replacement customer generally costs more than retaining an existing one. This analysis combines data preparation, K-Means segmentation, and an artificial neural network (ANN) to identify customers who may require timely retention support.

The dataset contains 7,043 customers, of whom 1,869 churned. The overall churn rate is **26.54%**. The clearest observed risk signal is early tenure: customers with 0–12 months of tenure churned at **47.44%**, compared with **9.51%** among customers with 49–72 months of tenure. Customers with monthly charges above 70 also showed elevated churn at **35.15%**.

The ANN achieved **78.14% accuracy** and a **0.8075 ROC-AUC** on the held-out test set. Its recall was **45.99%**, meaning the model identified fewer than half of the customers who churned. It should therefore be used as a prioritization aid, not as an automatic decision-maker.

Recommended management priorities

1. Create a first-year onboarding and intervention journey for new customers.
2. Review high-charge accounts for plan fit, service issues, and retention offers.
3. Provide tailored support for senior customers and customers without dependents.
4. Use churn probabilities to rank outreach, while incorporating customer value and contact history.
5. Validate every intervention through controlled experiments and monitor retention uplift.

1. Business Problem and Requested Outcome

The business problem is customer churn: customers who discontinue their service reduce recurring revenue and create additional acquisition and service costs. The organization needs an evidence-based way to understand which customers are more likely to leave, which groups share similar characteristics, and where retention teams should focus their effort.

The requested outcome was a complete churn-analysis workflow rather than a single model. The project needed to:

- prepare reliable customer data for analysis;
- identify meaningful customer segments;
- build and evaluate a churn-prediction model;
- translate the results into practical retention recommendations; and
- provide reusable outputs that can be reviewed and handed to business and technical stakeholders.

Success therefore means more than achieving a high model score. The work must connect data quality, analytical evidence, predictive performance, business action, and measurement into one traceable process.

2. Business Question and Scope

The central question is: **Which customer characteristics and segments are associated with churn, and how can the business prioritize retention activity?**

The work delivers three connected business outputs:

- Data preparation: A consistent, encoded, and scaled analytical dataset with reproducible train-test artifacts.
- Customer segmentation: K-Means clusters that group customers with similar characteristics and churn behavior.
- Predictive modeling: An ANN that estimates churn probability for held-out customers.

The results describe associations in a historical snapshot. They do not prove that a particular customer characteristic causes churn, and they should not be interpreted as guaranteed individual outcomes.

3. Analysis Approach

The analysis followed a staged workflow:

6. **Data preparation:** inspect the source data, standardize schema and types, handle potential missing values, encode categorical fields, scale numeric fields, and create a stratified train-test split.
7. **Exploratory analysis:** calculate the overall churn baseline and compare churn rates across tenure, charges, demographic, service, and contract groups.
8. **Customer segmentation:** apply K-Means to the encoded and scaled feature space, use the elbow method to compare candidate cluster counts, and profile the resulting four clusters.
9. **Predictive modeling:** train an ANN on the prepared training data, use validation and early stopping during training, and evaluate on the held-out test set.
10. **Business translation:** combine descriptive findings, cluster profiles, and churn probabilities into prioritized retention actions and a measurement plan.

This sequence separates preparation from analysis, analysis from prediction, and prediction from business action. It also makes clear which conclusions come from descriptive comparisons, which come from clustering, and which come from model evaluation.

4. Dataset and Data Preparation

The source data contains demographic, service, commercial, and target fields:

Category	Fields
Demographic	gender, SeniorCitizen, Dependents
Service	PhoneService, MultipleLines, InternetService
Commercial	Contract, MonthlyCharges, tenure
Target	Churn

The preparation workflow trimmed column and text-field whitespace, converted numeric fields to numeric types, encoded categorical variables with one-hot encoding, and standardized numeric fields with `StandardScaler`. The current source snapshot contained no missing values. Median and mode imputation logic was retained as a defensive pipeline step so the workflow can handle future incomplete batches consistently.

The data was split using an 80/20 stratified split. Features and labels were saved separately as `X_train.csv`, `X_test.csv`, `Y_train.csv`, and `Y_test.csv`. The fitted preprocessing object was saved as `preprocessing_pipeline.joblib` to support repeatable transformation.

5. Exploratory Findings

The dataset contains 5,174 retained customers and 1,869 churned customers.

Measure	Value
Total customers	7,043
Churned customers	1,869
Retained customers	5,174
Overall churn rate	26.54%

3.1 Key segment comparisons

Attribute	Segment	Customers	Churn rate
SeniorCitizen	0	5,901	23.61%
SeniorCitizen	1	1,142	41.68%
Dependents	No	4,933	31.28%
Dependents	Yes	2,110	15.45%
Contract	Month-to-month	3,875	27.33%
Contract	One year	1,473	25.80%
Contract	Two year	1,695	25.37%
InternetService	DSL	3,947	27.41%
InternetService	Fiber optic	3,096	25.42%

3.2 Tenure and monthly charges

Tenure group	Customers	Churn rate
0–12 months	2,186	47.44%
13–48 months	2,618	23.64%
49–72 months	2,239	9.51%

Monthly charge group	Customers	Churn rate
Up to 35	1,759	11.26%
36–70	1,793	24.76%
Above 70	3,491	35.15%

The strongest practical implication is that the first year of the customer relationship deserves disproportionate attention. Charge level is also a useful prioritization variable, especially when combined with tenure and service experience.

6. Customer Segmentation

K-Means clustering was applied to the encoded and scaled feature space. The elbow method evaluated cluster counts from 2 through 8. Four clusters were selected as an operational balance between useful differentiation and manageable campaign complexity.

Cluster	Customers	Churn rate	Avg. tenure	Avg. monthly	Business

				charges	interpretation
0	1,979	10.71%	29.33	25.77	Lower-charge, lower-risk customers
1	2,116	44.52%	12.55	73.20	Short-tenure, high-charge, high-risk customers
2	1,806	13.23%	58.35	88.07	Long-tenure, high-charge, lower-risk customers
3	1,142	41.68%	33.30	79.82	Senior-customer segment with high churn risk

Cluster 1 is the clearest retention priority because it combines short tenure, high charges, and the highest churn rate. Cluster 3 also warrants targeted support. Cluster 2 demonstrates that high charges alone do not imply high churn: established relationships can materially reduce risk.

7. Predictive Modeling

The predictive model is a binary-classification ANN implemented with TensorFlow/Keras. It uses two hidden layers with ReLU activation, dropout regularization, and a sigmoid output layer. Binary cross-entropy is used as the loss function, Adam as the optimizer, and early stopping is used to reduce overfitting.

5.1 Test-set performance

Metric	Result	Interpretation
Accuracy	78.14%	Correct classification rate across the test set
Precision	61.87%	Share of predicted churners who actually churned
Recall	45.99%	Share of actual churners detected by the model
ROC-AUC	0.8075	Ranking quality across probability thresholds

ROC-AUC indicates useful ranking performance, but recall is comparatively modest. A retention team should therefore test different probability thresholds and assess the cost of missed churners against the cost of contacting customers who would have stayed.

8. Retention Strategy

Priority	Target	Recommended action	Success measure
1	New customers in months 0–12	Welcome journey, service education, early satisfaction check, issue resolution	First-year churn and onboarding completion
2	High-charge customers,	Plan-fit review, bundle	Churn reduction and

	especially Cluster 1	optimization, billing explanation, targeted offer	offer acceptance
3	Senior customers, especially Cluster 3	Simplified communications, priority assistance, proactive service follow-up	Retention and service satisfaction
4	Customers without dependents	Personalized value messaging and relevant support	Segment churn rate and campaign uplift
5	High-probability customers identified by ANN	Rank outreach using probability, customer value, and contact history	Precision, recall, uplift, and return on intervention

The model score should support human-reviewed prioritization. Before deployment, the business should define contact policies, avoid discriminatory treatment, protect customer privacy, and test whether interventions improve outcomes rather than merely identifying risk.

9. Limitations and Risk Controls

- The dataset is a static snapshot, so seasonal behavior and customer-level churn history cannot be assessed.
- Important operational variables are absent, including payment method, complaints, support contacts, outages, competitor pricing, and customer lifetime value.
- The ANN is less interpretable than simpler models. Logistic regression and tree-based baselines should be evaluated before selecting a production model.
- Segment comparisons are observational. They identify patterns, not causal effects.
- Model performance should be monitored over time for drift, calibration, subgroup differences, and changes in recall.
- Retention campaigns should be evaluated with a control group and clear cost-benefit measures.

10. Reproducibility and Deliverables

The project includes the preprocessing notebook, transformed dataset, train-test feature and target files, serialized preprocessing pipeline, cluster assignments and profiles, trained ANN, predictions, metrics, and visualizations. These artifacts allow a supervisor, analyst, or project stakeholder to trace the workflow from source data through analysis outputs.

The principal outputs are:

- `preprocessed_dataset.csv` — encoded and scaled analytical dataset.
- `X_train.csv`, `X_test.csv`, `Y_train.csv`, `Y_test.csv` — stratified modeling split.
- `preprocessing_pipeline.joblib` — fitted transformation logic for reuse.
- `cluster_assignments.csv`, `cluster_profiles.csv` — segment membership and profiles.
- `ann_model.keras`, `ann_predictions.csv` — trained model and test predictions.
- `metrics_summary.txt` — recorded evaluation metrics.
- `elbow_method.png`, `clusters_pca.png`, `ann_training_history.png`, `confusion_matrix.png` — supporting visual evidence.

Conclusion

The analysis identifies a clear retention opportunity among early-tenure and high-charge customers, with additional targeted attention warranted for senior customers and customers without dependents. K-Means segmentation adds operational context, while the ANN provides useful but imperfect churn ranking.

The recommended next step is a measured pilot: prioritize the highest-risk, highest-value customers, assign interventions by segment, retain a control group, and evaluate incremental retention, customer experience, and financial return. This approach turns the analysis into a testable decision-support process rather than an unsupported automatic targeting system.Appendix: Supporting Visualizations

The following figures provide visual evidence for the segmentation and predictive modeling results discussed in this report.

Elbow Method

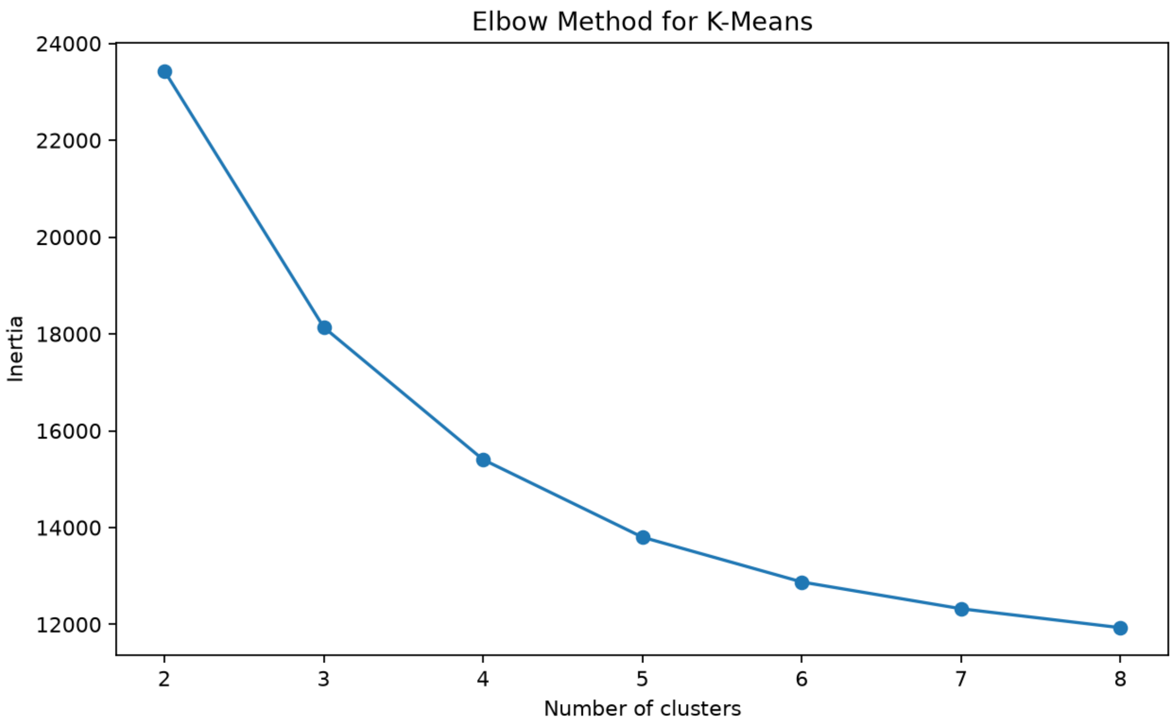

Figure: Elbow Method

Customer Segments PCA

Figure: Customer Segments PCA

ANN Training History

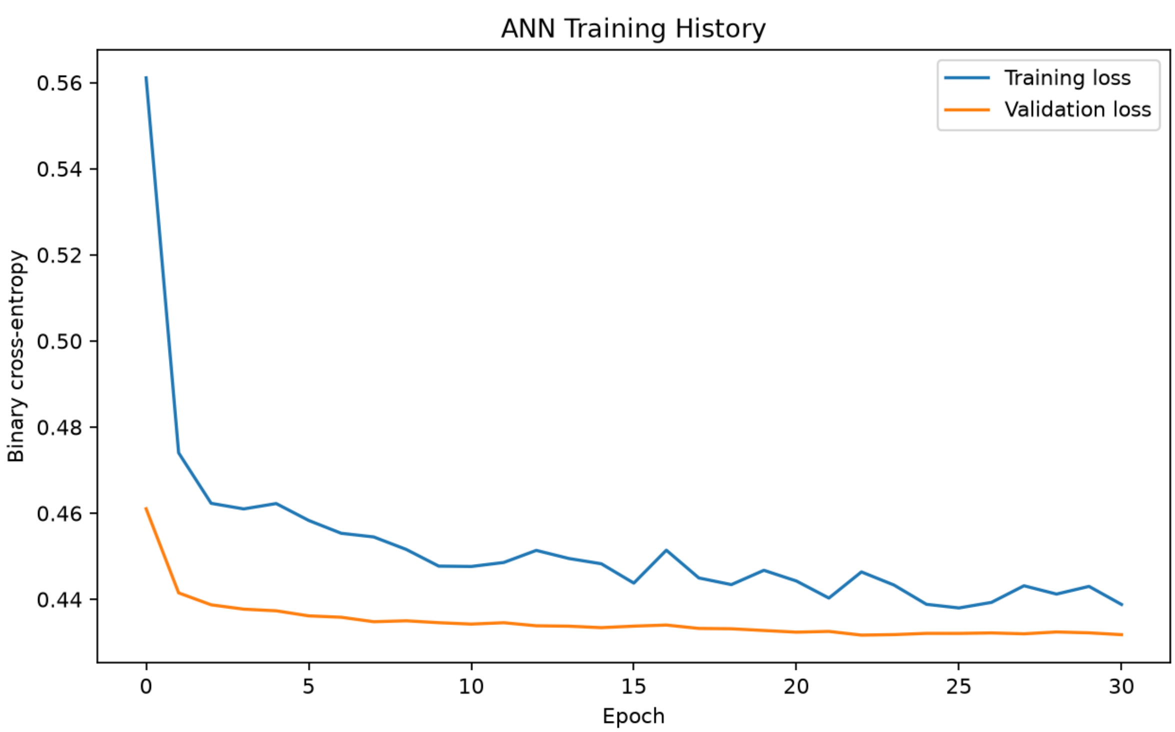

Figure: ANN Training History

ANN Confusion Matrix

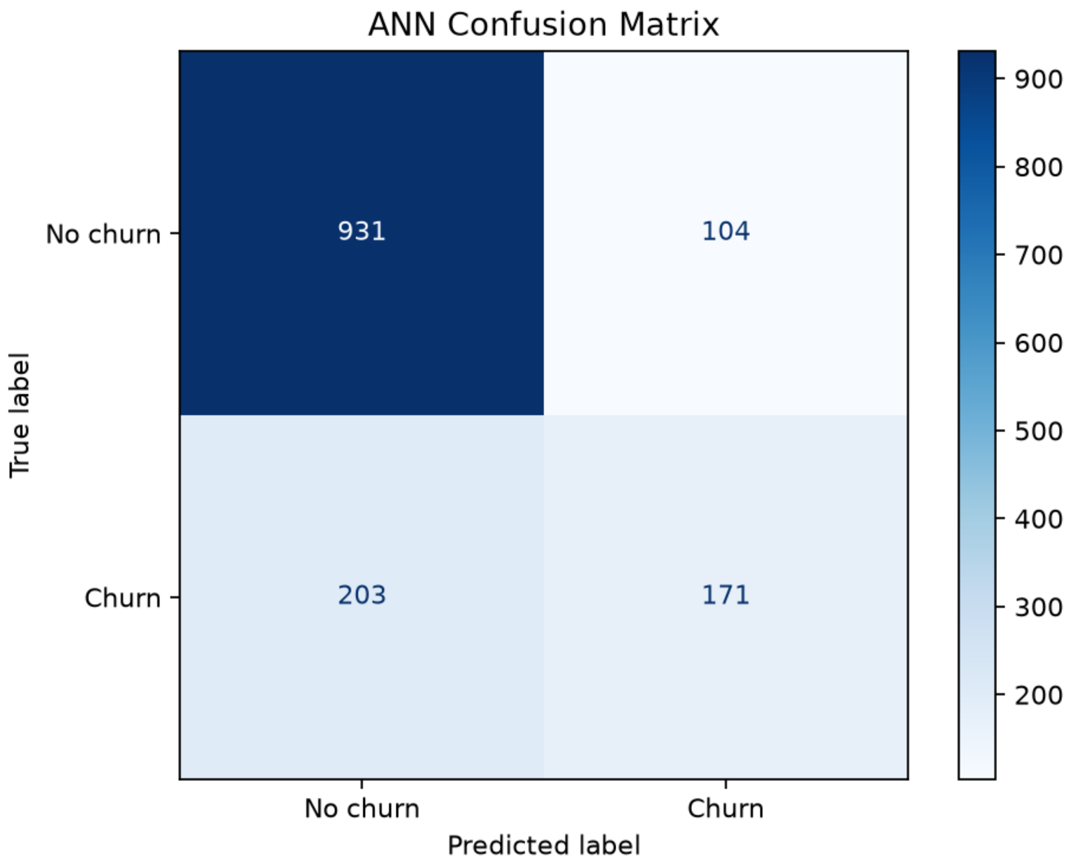

Figure: ANN Confusion Matrix